

SOFTWARE REQUIREMENTS SPECIFICATION

St George Hospital Management System

CENTRALISED • SECURE • MULTI-BRANCH • INSTALLABLE PWA

Document	Project / group	Version	Status
Assessment 3 — Final SRS	Project 20 • Group G3	Draft 1.0 • 5 Aug 2026	For team, supervisor and lecturer validation

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Document control and assumptions

Item	Baseline used in this draft
Solution form	Installable responsive Progressive Web Application using secure PHP/MVC services and MySQL.
Requirements	Client baseline FR1–FR48 and NFR1–NFR20 retain their identifiers. Proposed FR49–FR64 are not approved until written supervisor evidence is recorded.
System vs prototype	All 14 modules are specified in the target architecture; the student prototype may prioritise core end-to-end workflows and record deferrals transparently.
Data	Synthetic patient, clinical and payment data only during development and assessment.
External services	SMS, email, payment and AI use sandbox, mock or rule-based services until contracts, credentials and privacy review exist.
Legal context	Australian privacy law/APPs are primary. The release does not integrate with the national My Health Record. HIPAA is not claimed as governing Australian law.
Diagram maturity	The supplied diagrams are design drafts; the data dictionary and change register below extend omissions and identify required redraws.

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Executive summary

St George Hospital Group is looking for a single source of truth, as opposed to each location using its own, across Australian branches. Instead of email, paper and telephone-only booking, the proposed system will be a PWA that can be installed, secure PHP services, and a central MySQL database. This SRS preserves the client's FR1–FR48 and NFR1–NFR20, specifies the full 14-module target, and differentiates sixteen gap-closing candidate requirements which must be approved by the supervisor. There is measurable acceptance evidence related to architecture, data, interface, privacy, and testing and implementation decisions. Private sign-off, addressing critical security issues, a representative user acceptance test and the full evidence of contribution are also prerequisites for production readiness.

1. Introduction and project description

1.1 Business background and problem

St George Hospital has several super specialty hospitals in various states of Australia. Before this project, the client was using an outdated and decentralized healthcare management system. Based on the client's business, five concrete problem scenarios were identified as the obstacles to the client's business.

- **Data Sharing Challenges:** hospital files were sent and received by email in a manual process that often resulted in the loss of files in spam folders. The download of files to a local system before moving to a patient's folder created a complex data management cycle between branches.
- **Inefficient Appointment Booking:** patients were only able to book appointments by calling on hospital reception which made staff very busy during reception hours, and manually updating doctors' appointments.
- **Physical File Storage and Space Shortages:** patient records were kept on physical paper, and a new file was created for each visit, causing huge problems with space storage and retrieval of historical health records.
- **Lack of Secure Data Storage:** patient health records were stored in non-encrypted format, while every user of the system had full access to read and write data — which didn't adhere to Australian Privacy Principles (APPs) and NIST policies and put the hospital at risk of cyber-attacks.
- **Limited Accessibility:** Legacy system could only be used by on-site personnel to enter and send information to physicians and lab associates through email. There was no direct patient contact with their own health care management.

1.2 Project Purpose and Objectives

This project aims to create a centralized, mobile health care management system that addresses the shortcomings and weaknesses of the hospital's current system. The main business goals are to:

- Create a centralized, single point-of-entry healthcare system with all remote branches connected to and serviced from a single location at the hospital, thereby eliminating the use of email as a way to share data.
- Ensure that all medical records are digitized, so there is no use of paper within the facility and health records can be retrieved at a few seconds notice.
- Automated online scheduling solution that allows patients to book, view and manage appointments without a phone call to a doctor, due to availability.
- Design a strong role-based security system that limits access to patient information to authorized personnel, safeguarding the confidentiality, integrity and availability of patient information from cyber attacks.
- Provide direct communication across departments, multiple clinics/branches and secure access to all stakeholders (patients, doctors, hospital/clinic staff and admin) via a secure transmission channel.

1.3 Updated scope and deliverables

This project includes the Design, Development and Implementation of a native mobile application to be downloaded, installed and retained on users' mobile device until manually uninstalled, which has four primary panels (Doctor, Patient, Hospital/Clinic and Admin), with other role based panels also included (Branch Manager, Nurse, Receptionist and Lab Technician).

These are the key operational processes which will be automated:

- Data access control and management
- Organising and reporting through multiple branches
- Registration of patients and centralised medical records, including a global patient ID which is valid in all branches
- Doctor availability, conflict prevention, automatic reminder of appointments
- Staff and doctor scheduling, shift distribution and treatment reporting
- Caring for patients in hospital (admission and ward)
- Pharmacy and stock management, including stock alerts and purchase orders
- Laboratory and diagnostics – test requests, results, handling of reports
- Accounting, bookkeeping, reporting, and other financial aspects, such as itemized invoices, online payment gateways, etc.
- Reports and analytics, including dashboard-based statistics which can be exported.
- Ensuring communication and notifications through SMS/email.
- Audit and security logging
- Backing up and recovering data.
- Application of various tools to solve problems in the system

The scope includes protecting general client–server transmissions with HTTPS/TLS and protecting sensitive shared data with AES encryption via a public-key cryptosystem, in order to protect sensitive data.

1.4 Key Deliverables

The following artefacts will be produced on completion of this project:

- Software Requirement Specification (SRS) Report with complete requirement traceability, tracing Functional and Non-Functional Requirements back to Business Requirements.
- System Architecture & Data Models, Entity Relationship Diagrams (ERD), Class Diagrams and Data Flow Diagrams (DFD).
- User Interface (UI) Storyboards and Input/Output (I/O) form designs for the native mobile app.
- Data Protection Impact Assessment (DPIA) including data encryption and anonymisation methods.
- Project Management Artefacts, including a Work Breakdown Structure (WBS), Gantt chart and Individual Contribution Statement.
- System Test Plan and Implementation Plan.
- Working Prototype of the native mobile application.

1.5 Methodology, stakeholders and success

The team will use short Agile iterations with a prioritised backlog, weekly review and formal change control. Stable requirement identifiers, acceptance criteria and traceability follow ISO/IEC/IEEE 29148 principles (ISO/IEC/IEEE 2018). Trello should record owner, due date, acceptance criteria and status; GitHub should preserve individual commits and review; minutes should record decisions and blockers. Stakeholders are hospital management/IT, branch managers, clinical and administrative staff, patients, the supervisor and controlled service providers. Success is the verified client baseline, no unresolved critical privacy/security/data-loss defect, a restorable release and representative completion of core user journeys.

2. System architecture and environment

2.1 Architecture

The team will follow a series of short agile iterations, a prioritised backlog, weekly review and formal change control. The requirement identifiers, acceptance criteria and traceability are stable and meet the principles of ISO/IEC/IEEE 29148 (ISO/IEC/IEEE 2018). The owner should be recorded on Trello, the due date, the acceptance criteria and the status of the issue should be captured on Trello, the individual commits should be captured on GitHub, and the minutes should capture decisions and blockers. Stakeholders are hospital management/IT, branch managers, clinical and administrative staff, patients, the supervisor and controlled service providers. Success is defined as the verified client baseline, no unresolved critical privacy/security/data-loss defect, a restorable release and representative completion of core user journeys.

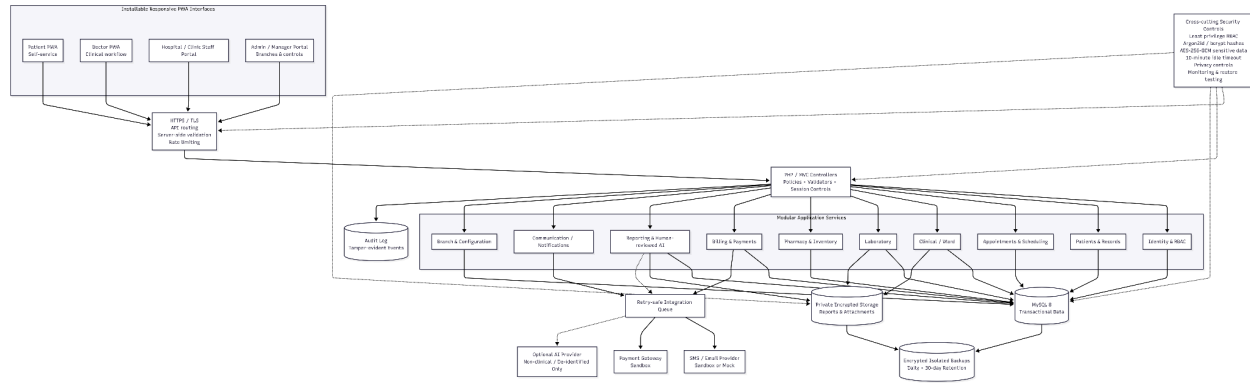


Figure 1. Proposed layered architecture and cross-cutting controls.

2.2 Development and deployment environment

Area	Recommended baseline
Application	Supported PHP 8.x release; Composer; Laravel or equivalent modular MVC framework; HTML5/CSS/JavaScript; accessible component library.
Data / web	MySQL 8.x; Apache or Nginx; ORM/prepared statements; versioned migrations and seeders.
Quality / collaboration	GitHub, Trello, VS Code, PHPUnit, API client, Playwright/Cypress or equivalent, OWASP ZAP, browser accessibility tools.
Developer workstation	Recommended minimum 4-core CPU, 8 GB RAM, 10 GB free storage; modern browser.
Initial staging	Approximately 4 vCPU, 8 GB RAM and 100 GB encrypted storage; resize from load-test evidence.
Operations	Separate dev/staging/production configuration; managed secrets; TLS certificate; monitored queue; encrypted isolated backups.

2.3 Proposed file and folder structure

The installable PWA offers Patient, Doctor, Hospital/Clinic and Administration interfaces, while RBAC is able to distinguish Patient, Receptionist, Doctor, Nurse, Lab Technician, Pharmacist, Branch Manager and Administrator. All traffic is in HTTPS. Every request through PHP/MVC controllers goes through validation without any client side validation and goes to modular services. The approach to store normalised transactions in MySQL includes foreign keys, indexes and database transactions. Private reports are kept out of the public webroot. Adapters separate messaging, payment and optional AI providers to prevent failing to disrupt hospital records (queuing and retrying possible). The following services are implemented cross-cutting: security, audit, backup and monitoring.

2.4 Assumptions and Dependencies

The system assumes that the development and assessment will be based on synthetic patient and clinical and payment data only. Sandbox, mock or rule based implementations of external SMS, email, payment and AI services will be provided until appropriate credentials, contracts and privacy approval are available. The system also requires secure web infrastructure, a supported PHP environment, MySQL support and proper deployment configuration, HTTPS/TLS support and appropriate deployment configuration. Proposed enhanced requirements are subject to approval of supervisor or stakeholder and will not be adopted as part of the official baseline.

3. Business, user and system requirements

3.1 Product Perspective

The St George Hospital Management System is a new, self-contained product designed to fully replace the hospital's outdated, decentralised and paper-heavy legacy system.

The product is a centralised native mobile application, supported by an administrative web backend, unifying all super-specialty hospital branches across Australia. Architecturally, it operates as a multi-tier client-server application. The client side consists of native mobile interfaces distributed across four primary panels (Admin, Doctor, Patient and Hospital/Clinic), with role-based views additionally supporting Branch Manager, Nurse and Receptionist functions. The server side acts as a centralized database and processing hub located at headquarters, handling real-time requests, branch synchronisation and enforcement of AES encryption before sensitive data is written to the database.

3.2 Product Functions

The system provides a comprehensive suite of automated functions to streamline hospital operations, including:

- **User Authentication and Role Management:** registration, secure login and enforcement of Role-Based Access Control (RBAC) across six user roles (Admin, Branch Manager, Doctor, Nurse, Receptionist, Patient).
- **Multi-Branch Management:** centralised control over hospital branches, staff assignments and branch-specific statistics.
- **Patient Management:** digital patient registration, global patient ID generation across branches and digitised medical history storage.
- **Appointment and Scheduling:** mobile appointment booking based on gender-wise or specialty-wise doctor filtration, live doctor availability, appointment accept/reject workflow, and automated SMS/email reminders (including a 2-hour pre-appointment notification and cancellation alerts).
- **Staff and Doctor Management:** tracking of doctor schedules, shift allocations and attendance; doctors can search patient history via unique patient ID.
- **Clinical Operations:** managing in-patient admissions and ward assignments, pharmacy inventory, laboratory test requests and digital diagnostic report attachment.

- Billing and Payments: automated generation of itemised invoices and integration with online payment gateways.
- Reports, Analytics and Auditing: visual dashboards, AI-generated report insights, feedback sentiment analysis and comprehensive audit-trail generation for compliance.

3.3 User classes and principal use cases

The system supports four primary user classes, consolidating six operational roles from the brief.

User class	Role and characteristics
Patients (End Users)	Registering accounts, filtering and selecting doctors, booking/viewing appointments, viewing personal medical records, bills and lab reports, and leaving doctor reviews. Low to medium technical expertise — interface must be highly intuitive, mobile-responsive and accessible.
Doctors (Medical Professionals)	Viewing assigned schedules, accepting/rejecting appointment requests, updating patient treatment notes, and searching patient history by unique ID. Medium technical expertise — requires an efficient interface for quick record updates between consultations.
Hospital/Clinic Staff (Receptionists, Nurses, Pharmacists, Lab Technicians)	Managing appointment approvals, recording patient vitals, issuing medicines, managing branch inventory and uploading lab results. Medium technical expertise, high frequency of use — requires rapid data-entry forms and real-time feedback.
Administrators / Branch Managers	Full system control, including staff account management, branch configuration, analytical report generation and audit log monitoring. High technical expertise — requires data-rich dashboards and advanced filtering.

4. External Interface Requirements

4.1 User Interfaces

The system employs a User-Centred Design (UCD) methodology, ensuring the interfaces for all four panels (Admin, Doctor, Patient, Hospital/Clinic) are intuitive, responsive and adhere to mobile accessibility standards (NFR13, NFR15).

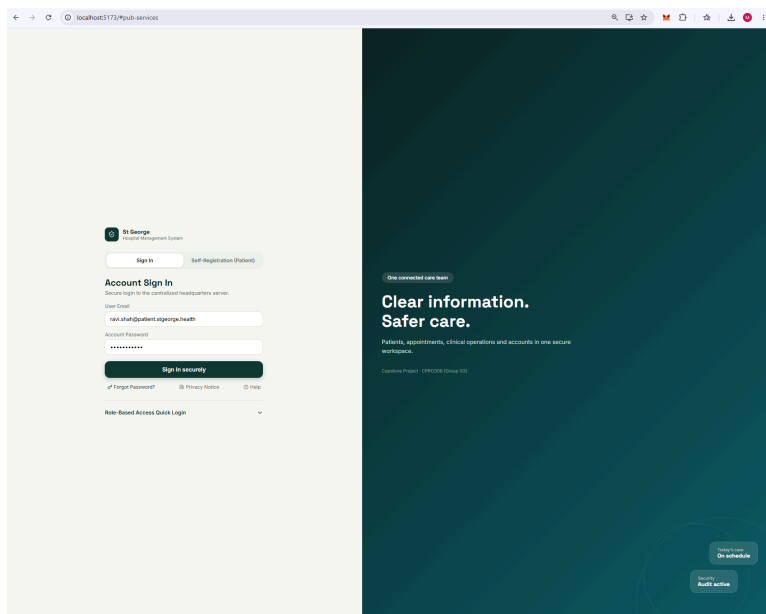
4.1.1 UI Design Methods and Concepts

- Consistency: standardised button placements (e.g. "Book Appointment", "Cancel", "Submit") and consistent colour schemes across all screens.
- Real-time Feedback: loading indicators during database queries, with immediate confirmation pop-ups when an appointment is accepted, rejected or cancelled.
- Error Prevention: input forms use dropdown menus (e.g. Doctor Specialty selection) and calendar pickers rather than free-text fields, supported by input validation and error messaging (NFR14).

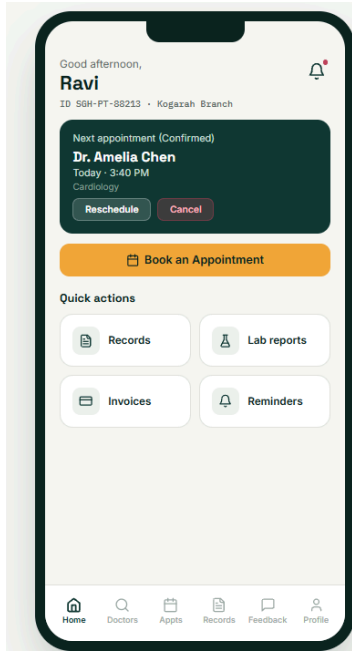
4.1.2 User Interface Storyboards & Layouts

The core user journey for the Patient panel follows this sequence:

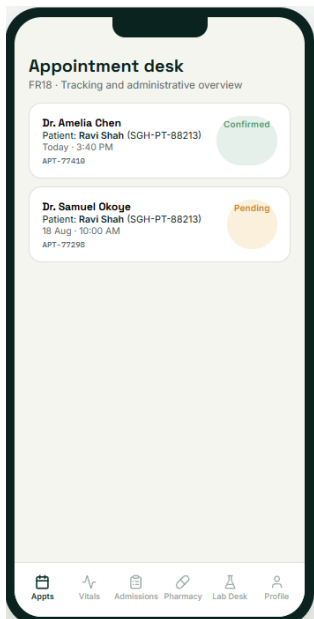
- Login/Register Screen — user authenticates with an encrypted password (FR3, NFR8).



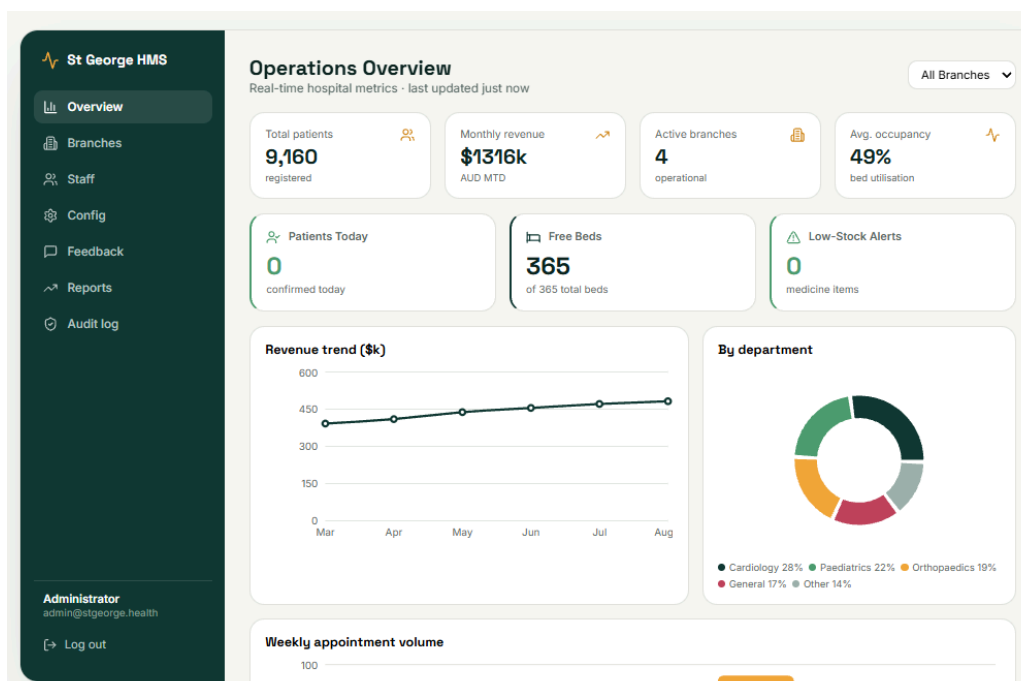
- Main Dashboard — displays upcoming appointments and a search bar.
- Filtration Screen — user filters available doctors by Gender and Specialty.
- Doctor Profile — displays the doctor's reviews, schedule and a "Request Appointment" button.
- Confirmation Screen — displays a success message once the doctor accepts the request, with automated notification triggered (FR20).



Whereas, receptionist dashboard and admin dashboard looks like the image below:



Receptionist Dashboard



Admin Dashboard

4.1.3 Input Data Forms

- Patient Registration Form: First Name, Last Name, Email, Password, Date of Birth, Gender, Contact Number, Pre-existing Medical Allergies.
- Appointment Request Form: Select Branch, Select Specialty, Select Doctor, Preferred Date (calendar picker), Preferred Time Slot (dropdown).

4.1.4 Output Report Forms

- Patient Invoice Report: dynamically generated PDF showing patient ID, itemised consultation and medication charges, and payment status (Paid/Pending) (FR37, FR38).
- Branch Manager Statistical Report: visual dashboard and exportable Excel/PDF sheet detailing weekly patient volume, revenue and pharmacy inventory alerts (FR41, FR42).

4.2 Hardware Interfaces

- Client-Side Hardware: Patient and Doctor panels require smartphones (Android or iOS) with touchscreens and network connectivity. Admin and Reception panels require standard desktop or laptop computers with keyboard and mouse input for rapid data entry.
- Server-Side Hardware: the centralised backend will be hosted on cloud-based server infrastructure. As the brief leaves specific hardware to the development team's discretion, the team recommends a minimum of a quad-core CPU (2.5GHz+), 8GB RAM and SSD storage (100GB+) to support responsive database retrieval under load — this should be validated during implementation planning rather than treated as a fixed client requirement.

- **Networking Hardware:** all client devices require active Wi-Fi or cellular connectivity to maintain a continuous connection with the central headquarters' server.

4.3 Software Interfaces

4.3.1 Software Development Environment

- **Frontend Mobile Application:** the team recommends a cross-platform framework (e.g. Flutter or React Native) to allow consistent deployment on both iOS and Android app stores — a technology choice made by the team, not specified by the client.
- **Backend Environment:** a secure LAMP-style stack (Linux, Apache, MySQL, PHP), with business logic and APIs handled server-side.
- **Database Management:** MySQL as the central relational database, with indexing configured to optimize retrieval of patient and appointment records (NFR3).

4.3.2 Files and Folder Structure

Following the Modular MVC architecture (NFR19):

- **Models-** data schema and database communication logic (e.g. PatientModel, AppointmentModel).
- **Views -** UI components and dashboard layout screens.
- **Controllers -** business logic and input routing (e.g. AuthController, BillingController).
- **config** — stores encryption key references and database connection strings.

4.3.3 External APIs

- **Financial Interface:** integration with a payment gateway API (e.g. Stripe or PayPal) for electronic transactions (FR40).
- **Communication Interface:** integration with an SMS/Email gateway API (e.g. Twilio) to dispatch automated appointment reminders, including the 2-hour pre-appointment notification (FR20).

4.4 Communications Interfaces

- **Client-Server Protocol:** all data transmission between the mobile application and backend server occurs over HTTP and HTTPS (NFR9).
- **Data Exchange Format:** the team's recommended architecture uses RESTful APIs exchanging JSON payloads for efficient data loading (e.g. populating the doctor list).
- **Security Protocol:** all communication channels are secured using SSL/TLS. Sensitive health data payloads are additionally encrypted using AES with a public-key cryptosystem before transmission, consistent with the project's security-by-design approach (Section 1.5).

4. Functional and non-functional requirements

4.1 Baseline and prioritisation

Appendix A preserves client FR1–FR48 by identifier and states a test-oriented interpretation. Must requirements form the core acceptance baseline; Should/Could labels guide prototype sequencing but do not erase client scope. Appendix B preserves NFR1–NFR20 with measurable verification. FR46–FR48 are constrained to non-clinical, human-reviewed use and must not send identifiable health information to an external AI provider without approval.

4.2 Proposed enhancements

The client asks the team to research 15–20 additional FRs and obtain approval before finalising them. Appendix C proposes FR49–FR64 to close explicit gaps in inpatient/ward, audit/security, backup/recovery, configuration, consent and cross-branch control. Every row is marked Pending approval; the team must add supervisor evidence and an approval date before calling it baseline.

5. System and data design

5.1 Data flow and behavioural design

The Level 0 DFD treats the platform as one controlled process exchanging registration, appointment, clinical, billing and notification information with authorised actors and a payment gateway. The Level 1 draft decomposes authentication, appointments, patient records, pharmacy/inventory, laboratory, billing/payment and reporting processes. It is conceptually useful but too dense at its current scale; Appendix F therefore labels it a draft and a required redraw. The sequence diagram demonstrates authentication, booking, consultation and payment, but these should become four scenario-specific diagrams so alternatives and security boundaries are testable.

5.2 Database design and integrity

The supplied ERD establishes a core transactional subset: Branch, User, Patient, Doctor, Appointment, Lab Test, Medicine, Prescription and Invoice. The final logical model extends it with roles, staff, schedules, encounters/notes, ward/bed movement, lab order/result separation, inventory movements, payment, consent, notification, audit and backup entities. Appendix D defines keys and controls. Foreign keys prevent orphan records; transactions protect booking, dispensing and payments; versioned clinical entries prevent silent overwrite; and indexes cover global patient ID, doctor/time, branch/status and patient/date queries.

Database control	Design rule / acceptance evidence
Confidentiality	Encrypt sensitive fields/files using authenticated encryption; keys are separate; every protected read is authorised and auditable.
Integrity	PK/FK/unique/check constraints, server validation and atomic transactions; reconcile stock and payment balances.
Availability	Daily encrypted backups, 30-day retention, monitored failure and timed restore to staging.

Database control	Design rule / acceptance evidence
Data quality	Global patient identity, possible-duplicate review, controlled corrections, required provenance and timestamps.
Retention	Purpose-specific approved schedule, legal hold where applicable, secure deletion/anonymisation and disposal evidence.

5.2 Safety and system protection

The platform implements defense-in-depth to secure patient data against unauthorised modification or accidental destruction. Governance is achieved through server-side **role-based access control**, encrypted daily backups, and validated recovery protocols to mitigate operational risks. Release readiness is conditional upon resolving all critical privacy, data-loss, or core-workflow defects.

5.5 Governance and operational rules

System interactions are constrained by authorised branch scope and assigned duties. Clinical records are accessible only to verified personnel, preserving a globally unique patient identity and a traceable audit trail of significant changes. Financial reconciliation occurs without storing full card details, while external AI remains human-reviewed and non-clinical. Proposed enhancements **FR49–FR64** are not approved until written stakeholder evidence is recorded.

6. Security, ethics and Privacy/Data Protection Impact Assessment

6.1 Security and anonymisation design

Passwords are one-way hashed with Argon2id or bcrypt; they are never reversibly encrypted. TLS protects data in transit. Sensitive fields and private files use authenticated AES-256-GCM encryption, with keys stored separately and rotated through an approved secret/key service. AES is symmetric; public-key cryptography may wrap/exchange a key but is not an 'AES public key' (NIST 2023). Server-side RBAC, prepared statements/ORM, output encoding, CSRF protection, secure cookies, rate limits, malware checks, audit alerts and isolated backups support defence in depth. OWASP ASVS provides testable security verification (OWASP Foundation 2025).

Development uses synthetic data only. Analytics extracts remove direct identifiers, replace patient IDs with tokens, separate re-identification keys, suppress small groups and retain only necessary fields. No identifiable health information is sent to external AI without vendor/privacy approval. AI outputs are advisory, traceable, human-reviewed and never used for autonomous diagnosis or treatment.

6.2 Impact assessment

Privacy / ethical risk	Initial	Mitigation and evidence	Residual
Unauthorised viewing or change	Extreme	Least privilege; assigned-patient rules; privileged MFA proposal; encryption; negative access matrix; audit alerts	Medium
Excessive collection / indefinite retention	High	Purpose-specific fields; notices/consent; approved retention; access/correction; disposal log	Medium
Cross-branch or duplicate mismatch	High	Global ID; branch scope; identity checks; duplicate review; controlled correction	Medium
Vendor, report or AI disclosure	High	Minimisation; de-identification; contracts/location review; mock services; human review	Low–Medium
Staff curiosity / emergency misuse	High	Need-to-know access; break-glass reason/time limit; alert and retrospective review	Medium
Backup loss / ransomware	High	Encrypted isolated backups; monitored daily jobs; 30-day copies; restore drills	Low–Medium
Biased or misleading AI output	High	No clinical decisions; show source aggregates; retain feedback text; test error/bias; escalation	Medium

The privacy owner must validate data flows, lawful purpose, notices/consent, recipients, retention, access/correction, vendors and residual risks using the OAIC PIA method (OAIC 2021). Review is repeated after material scope, vendor, data or law changes. Unresolved High/Extreme residual risk blocks production. The Privacy Act and APPs are the primary context; national My Health Record obligations require separate assessment only if that integration is later approved.

6. Quality and Compliance Controls

The platform baseline necessitates maintaining stable requirements-to-test traceability, versioned technical documentation, and verifiable **Trello/GitHub** audit evidence. Specification compliance further demands WCAG accessibility verification, formal privacy-by-design reviews, validated recovery protocols, and representative UAT scenarios alongside individual contribution logs. Final release authorization remains conditional upon successful security drills, **controlled residual privacy risk**, and the resolution of all outstanding supervisor or lecturer feedback items.

7. User interface design

Interfaces use consistent navigation, plain language, role-specific dashboards, visible status, responsive layouts and recovery guidance. Forms identify required fields, validate inline and avoid sensitive error detail. Destructive actions require confirmation; tables provide search, filters and pagination; reports state branch/date scope and provenance. Core workflows target WCAG 2.2 Level AA, including keyboard operation, labels, contrast, focus visibility, error identification and accessible authentication (W3C 2024).

Interface	Inputs	Outputs and safeguards
Secure sign-in	Username/email; password; reset request	Generic failure; idle timeout; rate limiting; accessible labels; no health data cached
Patient dashboard	Profile, filters, appointment request, feedback	Next appointment, own released records/results/invoices, notifications; branch/doctor availability
Clinical workspace	Vitals, findings, notes, prescriptions, lab request	Allergy/history summary, signed versioned notes, orders; assigned-patient control
Lab/pharmacy	Result/report; dispense quantity; batch movement	Verification status, patient release, stock/expiry/allergy warnings and audit
Admin/manager	Branch/role/settings filters; report parameters	Scoped metrics, configuration confirmation, audit events and accessible exports

Appendix F includes a four-screen storyboard for sign-in, the patient dashboard, appointment booking and the operations dashboard. The storyboard uses synthetic placeholders and must be validated with representative patients and staff before final build decisions.

8. Project planning and team management

Member	Primary role	Owned evidence
Priya Paneru	Project Manager	Scope/schedule/risk/decision logs; submission and minutes
Usha Tamang	Systems Analyst	Requirements, traceability, use cases, DFD review
Manish Das	Full-stack Developer	PHP/API, database, deployment and security commits
Oyndrila Oyshi Saha	UI/UX & Frontend	Storyboards, accessibility and frontend commits
Krishala Niroula	QA / Tester	Test cases, defects, regression and acceptance evidence

8.1 Work breakdown and milestones

WBS	Work package	Owner(s)	Timing / exit
1	Initiation, stakeholder/problem analysis, charter	PM + Analyst	W1–W3 • Interim SRS reviewed
2	Requirements, architecture, data/UI models, privacy plan	Analyst + UI + Dev + QA	W2–W5 • Final SRS baseline
3	Core platform and end-to-end feature implementation	Developers	W4–W9 • demonstrable core journey
4	Integration, security, accessibility, performance and UAT	QA + all	W7–W11 • no critical defect
5	Deployment rehearsal, guides, handover and demo	All	W10–W12 • accepted demonstration

Activity	W3	W4	W5	W6	W7–8	W9	W10	W11	W12
Requirements/SRS	•	•	◆	◆					

Activity	W3	W4	W5	W6	W7–8	W9	W10	W11	W12
Architecture/data/UI	•	•	•	◆	•				
Implementation		•	•	•	•	◆	•	•	
Testing/privacy	•	•	•	•	•	•	•	◆	
Docs/deployment/demo				•	•	•	•	•	◆

◆ marks the assessment or exit milestone: Interim Week 3, Final SRS Week 5, presentation Week 6, mid-project deliverable Week 9, end-project deliverables Week 11 and formal demonstration Week 12. Every active Trello card must have owner, date, acceptance criteria and evidence; weekly iteration plans/reports, GitHub commits/reviews and meeting actions form the contribution record.

9. Test and implementation plan

Test level	Representative pass condition
Unit	Validators, permission rules, charges, schedules and stock rules return expected results.
Integration	Booking writes one appointment, rejects conflicts and queues one notification; lab/dispense/payment transactions reconcile.
Database	Constraints, transactions, indexes, migrations and rollback behave correctly on synthetic data.
Security	Unauthorised role matrix denied; injection/XSS/CSRF/session/file tests pass; secrets absent from source; ASVS evidence retained.
Performance	NFR1–NFR4 passed at 200 virtual users and agreed seeded volumes.
Accessibility/usability	Keyboard, screen-reader labels, contrast, errors and representative task completion meet agreed criteria.
Backup/recovery	Daily encrypted backup restores to staging within 30 minutes and reconciliation checks pass.
UAT	Representatives complete patient, consultation, lab, pharmacy, billing and reporting scenarios without critical defects.

Each test case records requirement ID, data, precondition, steps, expected/actual result, evidence, tester and defect link. Release is blocked by unresolved critical security, privacy, data-loss or core-workflow defects. Implementation uses separate development, staging and production configuration: freeze the approved release; back up the target; apply versioned migrations; configure secrets; deploy API/PWA; run smoke/security tests; seed only approved data; train pilot users; monitor availability, errors, login anomalies, queues and backups. Failed smoke tests trigger rollback to the prior release and the tested database recovery procedure.

10. Feedback, changes and individual evidence

Feedback / evidence issue	Required action	Status
No formal lecturer feedback in supplied source pack	Record every later comment with date, action, owner, evidence and disposition.	Awaiting team update

Feedback / evidence issue	Required action	Status
Platform conflict: installable mobile vs responsive web	Confirm PWA interpretation with supervisor and baseline one terminology.	Pending confirmation
Incomplete/overcrowded diagrams	Add missing roles/entities; redraw Level 1 DFD; split sequence scenarios.	Design change required
FR49–FR64 not approved	Obtain written supervisor/stakeholder decision; mark accepted/rejected/deferred individually.	Pending approval
Trello cards have no owners/dates/criteria; Testing/Done empty	Assign all work, link commits/tests/docs, move through workflow and post iteration forum links.	Evidence gap

Suggested improvements are a revised all-role use-case model, expanded logical ERD, readable process-specific DFDs/sequence diagrams, one requirements-to-design-to-test matrix, production cost/options analysis, and a formal privacy/security approval gate. The contribution statement in Appendix G must be completed with real links; raw Trello activity counts are not fair evidence of effort.

11. Conclusion

This SRS converts the client's fragmented and insecure multi-branch environment into a traceable specification for an installable responsive healthcare platform. It aligns the architecture, data model, interface, privacy controls, test strategy and implementation plan with the client baseline and measured quality targets. The specification is ready for team and supervisor validation, but not for an unqualified production claim: PWA interpretation, proposed requirements, lecturer feedback, diagram refinements and contribution evidence remain explicit approval actions. Pilot acceptance requires verified requirements, controlled residual risk, successful restore/security testing and representative UAT.

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Appendix A — Baseline functional requirements (FR1–FR48)

Source: Project 20 client brief. Verification language is a project interpretation; requirement IDs remain unchanged.

ID	Area	Requirement	MoSCoW	Verification
FR1	Authentication	Register users only after validating email, unique username and password.	Must	Positive/negative auth and permission tests; inspect audit/hash.
FR2	Authentication	Verify login credentials and reject invalid or unauthorised access with a generic error.	Must	Positive/negative auth and permission tests; inspect audit/hash.
FR3	Authentication	Hash passwords with bcrypt, Argon2id or an approved equivalent before storage.	Must	Positive/negative auth and permission tests; inspect audit/hash.
FR4	Authentication	Present a dashboard authorised for the authenticated user's role.	Must	Positive/negative auth and permission tests; inspect audit/hash.
FR5	Authentication	Allow administrators to add, modify and deactivate accounts, with actions audited.	Must	Positive/negative auth and permission tests; inspect audit/hash.
FR6	Branch	Allow administrators to create, update and deactivate hospital branches.	Must	Authorised CRUD, branch separation and reporting test.
FR7	Branch	Store a unique branch ID, name, location and contact details for every branch.	Must	Authorised CRUD, branch separation and reporting test.
FR8	Branch	Allow administrators to assign staff members to authorised branches.	Must	Authorised CRUD, branch separation and reporting test.
FR9	Branch	Maintain statistics and reports that can be filtered by branch.	Must	Authorised CRUD, branch separation and reporting test.
FR10	Branch	Allow branch managers to monitor admissions, discharges and resources for their branch only.	Must	Authorised CRUD, branch separation and reporting test.
FR11	Patient	Allow receptionists to register patients using validated demographic details.	Must	Patient lifecycle, own-record and role-scope test.
FR12	Patient	Assign one globally unique patient ID across all branches.	Must	Patient lifecycle, own-record and role-scope test.
FR13	Patient	Allow authorised staff to update only the patient fields permitted for their role.	Must	Patient lifecycle, own-record and role-scope test.
FR14	Patient	Preserve complete medical histories including diagnoses, allergies, prescriptions and treatment history.	Must	Patient lifecycle, own-record and role-scope test.
FR15	Patient	Allow authenticated patients to securely view the permitted portion of their own records.	Must	Patient lifecycle, own-record and role-scope test.
FR16	Appointment	Allow patients to request appointments with available doctors.	Must	Availability, conflict, status and notification integration test.
FR17	Appointment	Display current doctor availability by date and time.	Must	Availability, conflict, status and notification integration test.
FR18	Appointment	Allow receptionists to approve, modify or cancel appointments and record actors and reason.	Must	Availability, conflict, status and notification integration test.
FR19	Appointment	Prevent overlapping appointments for the same doctor using a transactional conflict check.	Must	Availability, conflict, status and notification integration test.

ID	Area	Requirement	MoSCoW	Verification
FR20	Appointment	Queue confirmation and reminder email or SMS after a valid appointment state change.	Must	Availability, conflict, status and notification integration test.
FR21	Staff	Allow administrators to add and update doctor and staff profiles and specialisations.	Must	Profile/schedule/assignment and clinical-role tests.
FR22	Staff	Store doctor schedules, visiting hours, shifts, leave dates and availability.	Must	Profile/schedule/assignment and clinical-role tests.
FR23	Staff	Allow doctors to access assigned patient files and append treatment notes.	Must	Profile/schedule/assignment and clinical-role tests.
FR24	Staff	Allow nurses to record authorised vitals and care observations.	Must	Profile/schedule/assignment and clinical-role tests.
FR25	Staff	Maintain staff attendance and shift-allocation records.	Should	Profile/schedule/assignment and clinical-role tests.
FR26	Pharmacy	Store medicines, quantities, expiry dates, minimum thresholds and suppliers.	Must	Stock, expiry, prescription and transaction reconciliation test.
FR27	Pharmacy	Notify pharmacists when available stock falls below its minimum threshold.	Must	Stock, expiry, prescription and transaction reconciliation test.
FR28	Pharmacy	Record every dispensed medicine, reduce stock and link it to a valid prescription and patient.	Must	Stock, expiry, prescription and transaction reconciliation test.
FR29	Pharmacy	Allow authorised users to generate purchase orders for required restocking.	Should	Stock, expiry, prescription and transaction reconciliation test.
FR30	Pharmacy	Block dispensing when medicine is expired or stock is insufficient.	Must	Stock, expiry, prescription and transaction reconciliation test.
FR31	Laboratory	Allow doctors to create laboratory or imaging test requests for patients.	Must	Order-result-release-search and file-authorisation test.
FR32	Laboratory	Allow lab technicians to record results and upload authorised reports to the relevant order.	Must	Order-result-release-search and file-authorisation test.
FR33	Laboratory	Notify the doctor and patient when a verified test result is released.	Must	Order-result-release-search and file-authorisation test.
FR34	Laboratory	Allow patients to securely view, download or print their released reports.	Must	Order-result-release-search and file-authorisation test.
FR35	Laboratory	Provide authorised search of test history by patient, branch, type, date and status.	Should	Order-result-release-search and file-authorisation test.
FR36	Billing	Calculate configured consultation, test and medication charges.	Must	Charge/invoice/payment/reconciliation test with sandbox gateway.
FR37	Billing	Generate an itemised invoice for each relevant visit or admission.	Must	Charge/invoice/payment/reconciliation test with sandbox gateway.
FR38	Billing	Allow patients to securely download or print their own invoices.	Must	Charge/invoice/payment/reconciliation test with sandbox gateway.
FR39	Billing	Record valid paid, pending or cancelled invoice status and audit its history.	Must	Charge/invoice/payment/reconciliation test with sandbox gateway.
FR40	Billing	Process electronic payment through a sandboxed gateway without storing full card details.	Should	Charge/invoice/payment/reconciliation test with sandbox gateway.

ID	Area	Requirement	MoSCoW	Verification
FR41	Reporting	Generate daily, weekly and monthly patient, revenue and appointment statistics.	Must	Scoped aggregate, timing, chart and export test.
FR42	Reporting	Allow administrators to export authorised reports as PDF and spreadsheet files.	Should	Scoped aggregate, timing, chart and export test.
FR43	Reporting	Display role-appropriate graphical dashboards summarising key metrics.	Should	Scoped aggregate, timing, chart and export test.
FR44	Reporting	Allow branch managers to compare authorised department performance.	Should	Scoped aggregate, timing, chart and export test.
FR45	Communication	Send and log approved appointment, result and billing notifications.	Must	Queue, delivery status, retry and audit test.
FR46	AI	Provide a non-clinical chatbot for supported booking, result-status, opening-hours and contact questions; escalate unsupported or clinical questions.	Should	Supported/unsupported, privacy, provenance and human-review test.
FR47	AI	Classify patient feedback as positive, neutral or negative while retaining the original comment for human review.	Could	Supported/unsupported, privacy, provenance and human-review test.
FR48	AI	Generate short summaries from validated monthly aggregates, show source values and require human review before external use.	Could	Supported/unsupported, privacy, provenance and human-review test.

Appendix B — Non-functional requirements (NFR1–NFR20)

ID	Quality	Measurable requirement	Verification
NFR1	Performance	Normal pages load within 3 seconds at the 95th percentile under the agreed network profile.	Timed browser test under agreed workload.
NFR2	Performance	Support at least 200 concurrent users without noticeable degradation.	200-user load test; <1% server errors and NFR1 maintained.
NFR3	Performance	Use database indexing for fast patient and appointment retrieval.	Explain plan/index inspection and representative query timings.
NFR4	Performance	Generate reports within 10 seconds for datasets under 10,000 records.	Timed report test with reproducible seeded dataset.
NFR5	Scalability	Add branches without structural database changes.	Create a branch through configuration/data only.
NFR6	Scalability	Support multi-server deployment if load increases.	Architecture review confirms stateless app nodes and external sessions/queues.
NFR7	Scalability	Maintain consistent performance as annual data volume grows.	Forecast-volume load test continues to meet NFR1–NFR4.
NFR8	Security	Store passwords using bcrypt, Argon2id or an approved equivalent.	Source/config review and hash inspection; no plaintext credentials.
NFR9	Security	Protect all client-server traffic with HTTPS/TLS.	TLS scan and HTTP-to-HTTPS redirect test.
NFR10	Security	Restrict sensitive modules with server-side RBAC.	The negative role-permission matrix denies every unauthorised route.

ID	Quality	Measurable requirement	Verification
NFR11	Security	End inactive authenticated sessions after 10 minutes.	Automated idle-session expiry test.
NFR12	Security	Log login attempts, data changes and deletions for auditing.	Audit entries show the actor, action, object, time and outcome.
NFR13	Usability	Provide an intuitive, consistent, mobile-responsive interface on major browsers.	Responsive/browser matrix and representative task completion.
NFR14	Usability	Provide validation, errors and guidance on all data-entry forms.	Positive/negative validation tests; no sensitive error disclosure.
NFR15	Accessibility	Support accessibility features including labels and sufficient contrast.	Core journeys tested to WCAG 2.2 AA using automated and manual checks.
NFR16	Availability	Achieve 99.5% uptime during agreed operational hours.	Monitoring report and availability calculation excluding approved maintenance.
NFR17	Reliability	Back up the database every 24 hours and retain copies for 30 days.	Scheduler/log inspection and retention inventory.
NFR18	Recovery	Recover from critical failure within 30 minutes of detection.	Timed restore exercise with reconciliation checks.
NFR19	Maintainability	Use modular MVC architecture.	Code and architecture review confirms separation and documented interfaces.
NFR20	Maintainability	Maintain installation, schema, administrator and change documentation.	Versioned docs are current for the release candidate.

Appendix C — Proposed enhanced requirements

ID	Gap area	Candidate requirement	Status
FR49	Consent	Record the privacy notice/consent version, purpose, grant/withdrawal time and evidence for each patient.	Pending approval
FR50	Security	Require multi-factor authentication for administrators and other privileged roles.	Pending approval
FR51	Security	Support emergency break-glass access only with a reason, time limit, alert and retrospective review.	Pending approval
FR52	In-patient	Admit, transfer and discharge a patient while preserving room/bed history and responsible staff.	Pending approval
FR53	In-patient	Display real-time bed availability and prevent double allocation.	Pending approval
FR54	In-patient	Record daily ward observations and generate an authorised discharge summary.	Pending approval
FR55	Audit	Provide authorised search and export of audit events by actor, branch, object, action and date.	Pending approval
FR56	Audit	Alert administrators to repeated failed logins, privileged changes and unusual record access.	Pending approval
FR57	Backup	Run configured encrypted backups, verify completion and alert on failure.	Pending approval
FR58	Backup	Allow an authorised administrator to initiate a controlled restore drill and record its result.	Pending approval

FR59	Configuration	Manage appointment durations, notification schedules, retention settings and feature flags through audited configuration.	Pending approval
FR60	Configuration	Maintain departments, wards, rooms, beds, service prices and inventory thresholds by branch.	Pending approval
FR61	Patient rights	Manage patient access and correction requests with identity verification, assignment, decision and audit trail.	Pending approval
FR62	Data quality	Detect possible duplicate patient records and require authorised review before merge.	Pending approval
FR63	Safety	Block or warn on known allergy and expired medicine conflicts before a prescription is dispensed.	Pending approval
FR64	Payments	Reconcile idempotent gateway callbacks and approved refunds without duplicate payment records.	Pending approval

Appendix D — Data dictionary

The physical schema must use consistent names/types, encryption annotations, indexes, retention classifications and migration versions.

Entity	Essential fields	Relationships / controls
Branch	branch_id PK; name; state; address; contact; status	One branch has users, appointments, inventory and reports.
Role	role_id PK; name; description	Many-to-many with User through UserRole; permissions are separate.
User	user_id PK; branch_id FK; email; username; password_hash; status; last_login	Authentication identity; never stores plaintext password.
UserRole	user_id FK; role_id FK; effective_from/to	Composite key prevents duplicate active assignments.
Patient	patient_id PK; user_id FK nullable; global_number; demographics; contact	Sensitive direct identifiers encrypted or tokenised where appropriate.
Staff	staff_id PK; user_id FK; branch_id FK; profession; registration_no; status	Supertype for doctor, nurse, technician and pharmacist profiles.
Schedule	schedule_id PK; staff_id FK; branch_id FK; start/end; type	Supports availability, shift and leave; indexed by staff/time.
Appointment	appointment_id PK; patient_id; doctor_staff_id; branch_id; start/end; status; reason	Transactional uniqueness/conflict rule for doctor/time.
MedicalRecord	record_id PK; patient_id FK; author_staff_id FK; record_type; content_ciphertext; created_at; version	Clinical entries are attributed and versioned, not silently overwritten.
Admission	admission_id PK; patient_id; branch_id; bed_id; admitted/discharged_at; status	Preserves bed movement and discharge history.
Bed	bed_id PK; branch_id; ward; room; number; status	Only one active admission allocation per bed.
LabOrder	order_id PK; patient_id; requester_id; test_type; status; requested_at	One order may have one or more verified results/documents.
LabResult	result_id PK; order_id FK; technician_id; result_ciphertext; file_uri; verified_at	File stored outside public web root; access is authorised and audited.
Medicine	medicine_id PK; name; batch; expiry; threshold; supplier_id	Stock is branch/batch aware; expired stock cannot be dispensed.

Entity	Essential fields	Relationships / controls
InventoryTransaction	transaction_id PK; branch_id; medicine_id; qty; type; actor_id; timestamp	Append-only movements produce traceable balance.
Prescription	prescription_id PK; patient_id; prescriber_id; issued_at; status	Items store medicine, dose, route, frequency and instructions.
Invoice	invoice_id PK; patient_id; visit/admission ref; total; status; issued_at	Line items explain every charge; total is derived/validated.
Payment	payment_id PK; invoice_id; gateway_reference; amount; status; received_at	No full card data; callback reference is idempotent.
Consent	consent_id PK; patient_id; purpose; notice_version; state; changed_at	Withdrawal does not erase the historical evidence.
Notification	notification_id PK; recipient; channel; template; related_ref; status; sent_at	Retry-safe queue; message content minimised.
AuditLog	audit_id PK; actor; role; action; object_type/id; outcome; timestamp; source	Append-only/tamper-evident; restricted retention and access.
BackupJob	job_id PK; type; started/ended; status; location_ref; verification	Tracks schedule, failure alerts and restore evidence.
Feedback	feedback_id PK; patient_id nullable; comment; sentiment; reviewed_by; created_at	Original text retained; model label is advisory.

Appendix E — Detailed use-case definitions

UC01 — Register and authenticate

Field	Definition
Primary actors	Patient (registration); all users (login)
Preconditions	Registration details are unique and valid, or an active account exists.
Main success flow	Validate input; create/locate account; verify password hash; establish secure session; load the authorised dashboard; log outcome.
Alternatives / exceptions	Reject duplicate/invalid registration, invalid credentials, inactive account or rate limit with a generic message.
Postcondition	An authorised session exists, or the failed attempt is logged without revealing account existence.
Traceability	FR1–FR5; NFR8–NFR12

UC02 — Manage users and branches

Field	Definition
Primary actors	Administrator; Branch Manager (read-only authorised scope)
Preconditions	Privileged users are authenticated and authorised.
Main success flow	Search object; validate requested change; apply within a transaction; refresh branch/role cache; write audit entry; show confirmation.

Field	Definition
Alternatives / exceptions	Deny unauthorized scope, duplicate branch identity or deactivation that would violate a dependency; provide corrective guidance.
Postcondition	The account/branch state is consistent and auditable.
Traceability	FR5–FR10; NFR5, NFR10, NFR12

UC03 — Register a patient

Field	Definition
Primary actors	Receptionist; Patient (self-service profile)
Preconditions	User is authorised; required identity and contact information is available.
Main success flow	Show privacy notice; validate minimum data; check possible duplicates; create a global patient ID; save record; confirm outcome.
Alternatives / exceptions	Potential duplicate is routed to authorised review; invalid data remains unsaved and fields are identified.
Postcondition	One patient record and consent/notice evidence exist; access is audited.
Traceability	FR11–FR15; proposed FR49, FR61–FR62

UC04 — Manage an appointment

Field	Definition
Primary actors	Patient; Receptionist; Doctor
Preconditions	The patient and doctor are active; the doctor has a published schedule.
Main success flow	Select branch, specialty, doctor and time; API rechecks availability; save request; authorised actor confirms/changes status; queue confirmation and reminder.
Alternatives / exceptions	Reject a conflicting slot and return alternatives; queue message retry without cancelling a valid appointment.
Postcondition	Exactly one auditable appointment exists in a valid state.
Traceability	FR16–FR20; NFR1, NFR3, NFR12

UC05 — Record consultation and ward observations

Field	Definition
Primary actors	Doctor; Nurse
Preconditions	An actor is assigned to the patient/encounter and permitted for the intended record type.
Main success flow	Review allergies/history; record findings or vitals; append treatment note; optionally create prescription/lab order; sign entry; audit access/change.
Alternatives / exceptions	Deny unassigned or wrong-role access; warn on allergy conflict; save an authorised draft where policy permits.

Field	Definition
Postcondition	Clinical entries are timestamped, attributed and versioned; previous signed content remains intact.
Traceability	FR14, FR23–FR24, FR31; proposed FR52–FR54, FR63

UC06 — Process a laboratory test

Field	Definition
Primary actors	Doctor; Laboratory Technician; Patient
Preconditions	An authorised order exists for the patient.
Main success flow	Technician accepts order; records result; uploads protected report; authorised verifier releases it; system notifies doctor/patient.
Alternatives / exceptions	Reject unsafe file or incomplete result; keep unverified result hidden from patient; queue notification retry.
Postcondition	Verified result is linked to the order and available only to authorised users.
Traceability	FR31–FR35; NFR9–NFR12

UC07 — Dispense medicine

Field	Definition
Primary actors	Pharmacist
Preconditions	Valid prescription and available non-expired branch stock exist.
Main success flow	Verify patient/prescription; check allergy, expiry and stock; record dispense; create inventory movement; trigger low-stock alert when required.
Alternatives / exceptions	Block expired/insufficient stock or unresolved allergy; refer correction to prescriber.
Postcondition	Dispense and stock balance are consistent, traceable and linked to the patient.
Traceability	FR26–FR30; proposed FR63

UC08 — Invoice and payment

Field	Definition
Primary actors	Billing Staff; Patient; Payment Gateway
Preconditions	An itemised unpaid invoice exists.
Main success flow	Send amount/reference to gateway; verify signed response; record status once; issue receipt; retain gateway reference without card details.
Alternatives / exceptions	Timeout leaves invoice pending until reconciliation; duplicate callback is idempotent; failure does not mark invoice paid.
Postcondition	Invoice/payment state is reconciled and auditable.
Traceability	FR36–FR40; proposed FR64; NFR9, NFR12

UC09 — Generate operations report

Field	Definition
Primary actors	Branch Manager; Administrator
Preconditions	The actor has reporting permission and an authorised branch scope.
Main success flow	Select metric/date/branch; query validated aggregate; display accessible chart/table; optionally export; log sensitive report access.
Alternatives / exceptions	Reject excessive/unauthorised scope; show data provenance and no-data state; keep AI summary unpublished until reviewed.
Postcondition	A reproducible, scoped report or export is produced within NFR4.
Traceability	FR41–FR44, FR48; NFR4, NFR10, NFR15

Appendix F — Design diagrams and storyboards

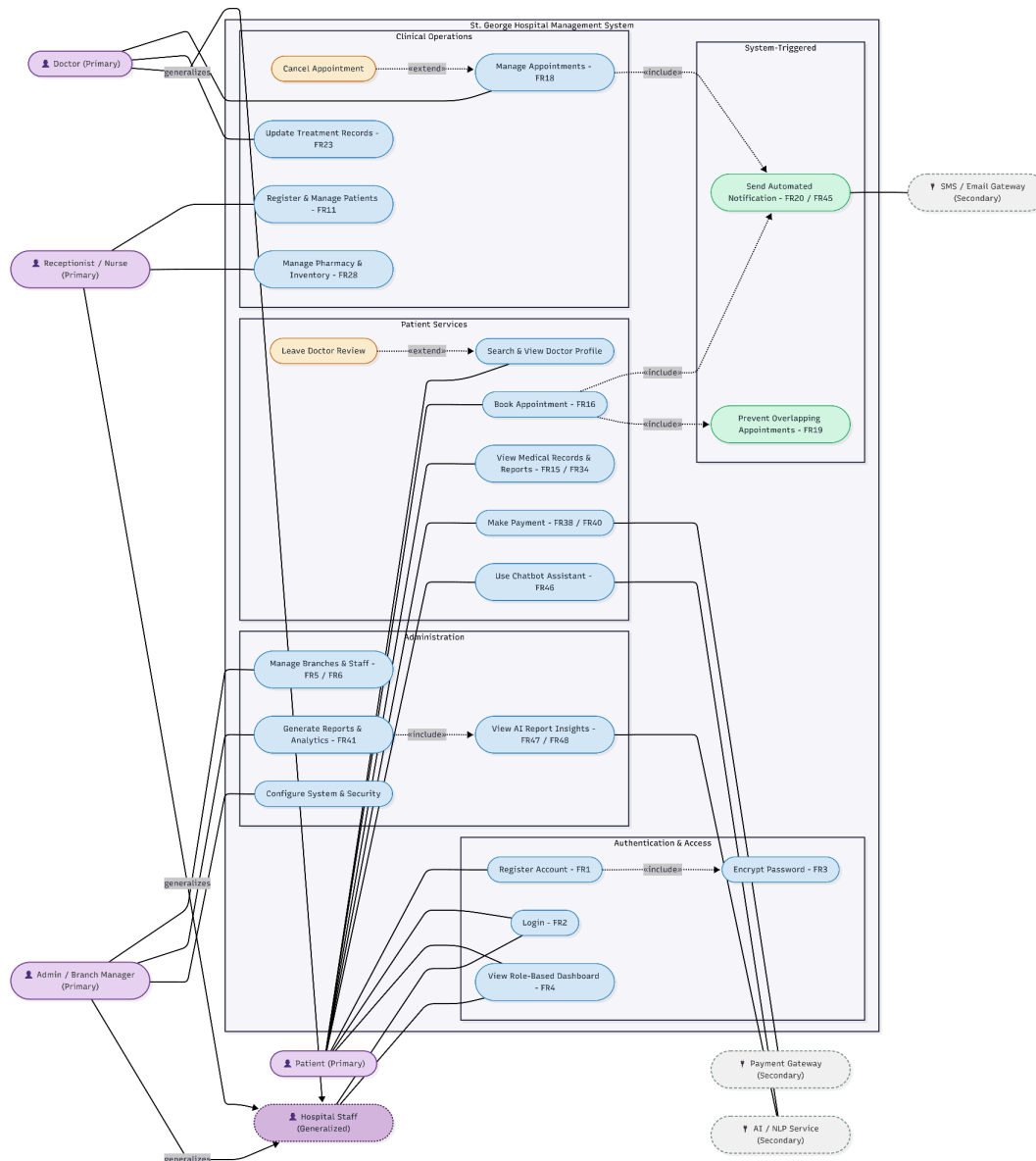


Figure F1. Supplied use-case diagram (draft; missing roles noted in Section 3)

Entity Relationship Diagram

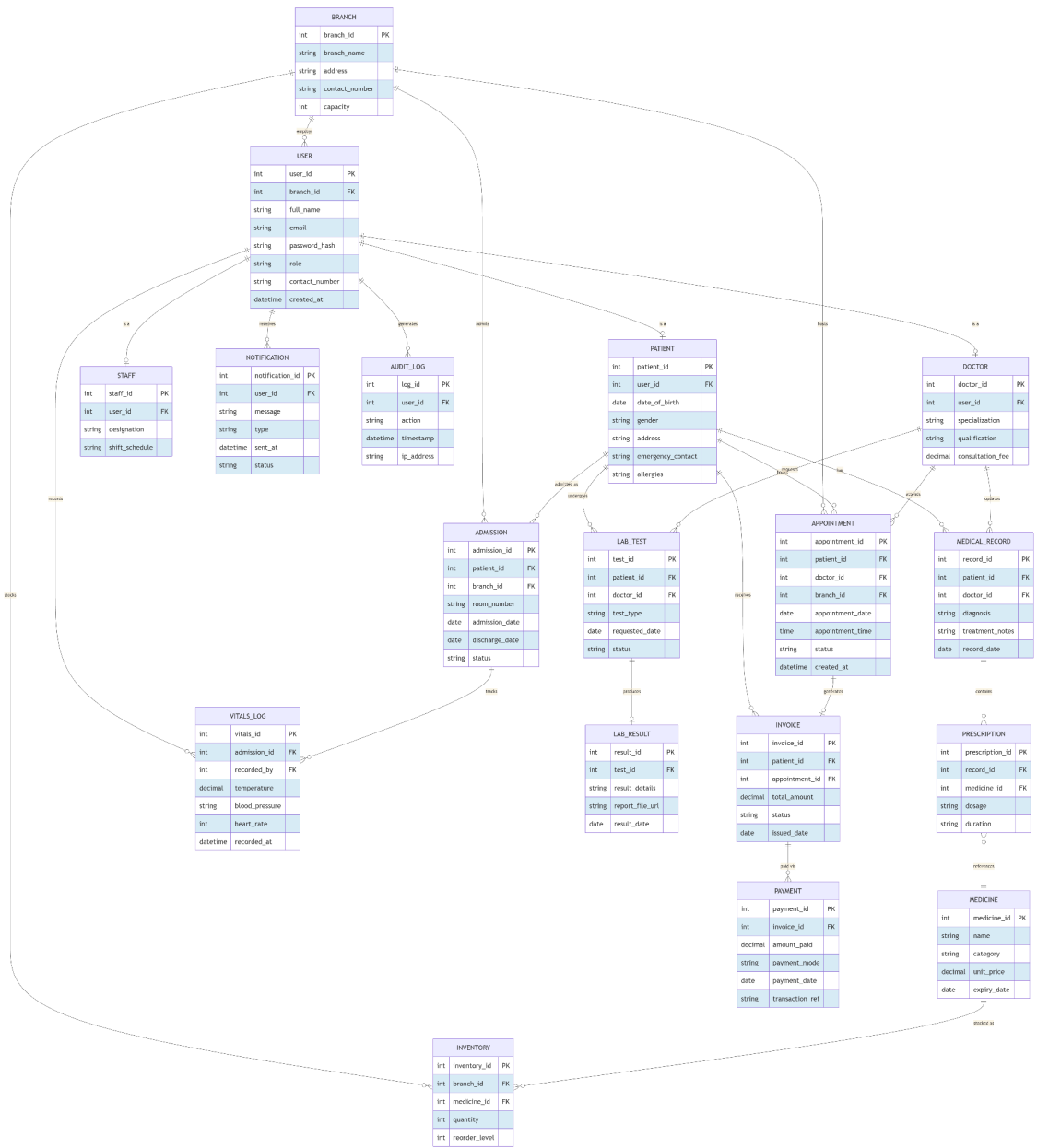


Figure F2. Supplied core entity-relationship diagram.

Class diagram

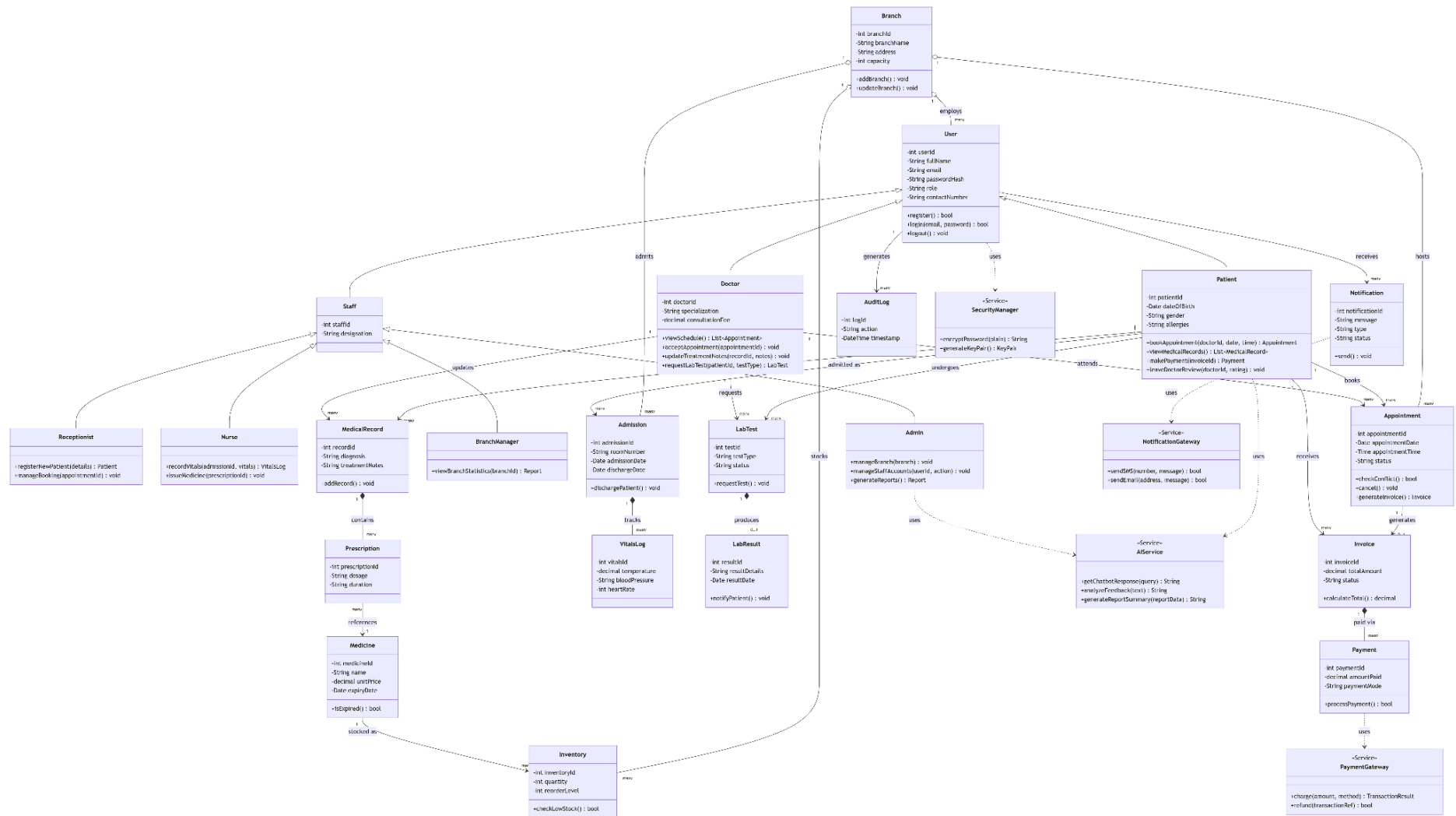


Figure F3. Supplied class diagram for the core object model.

0 DFD

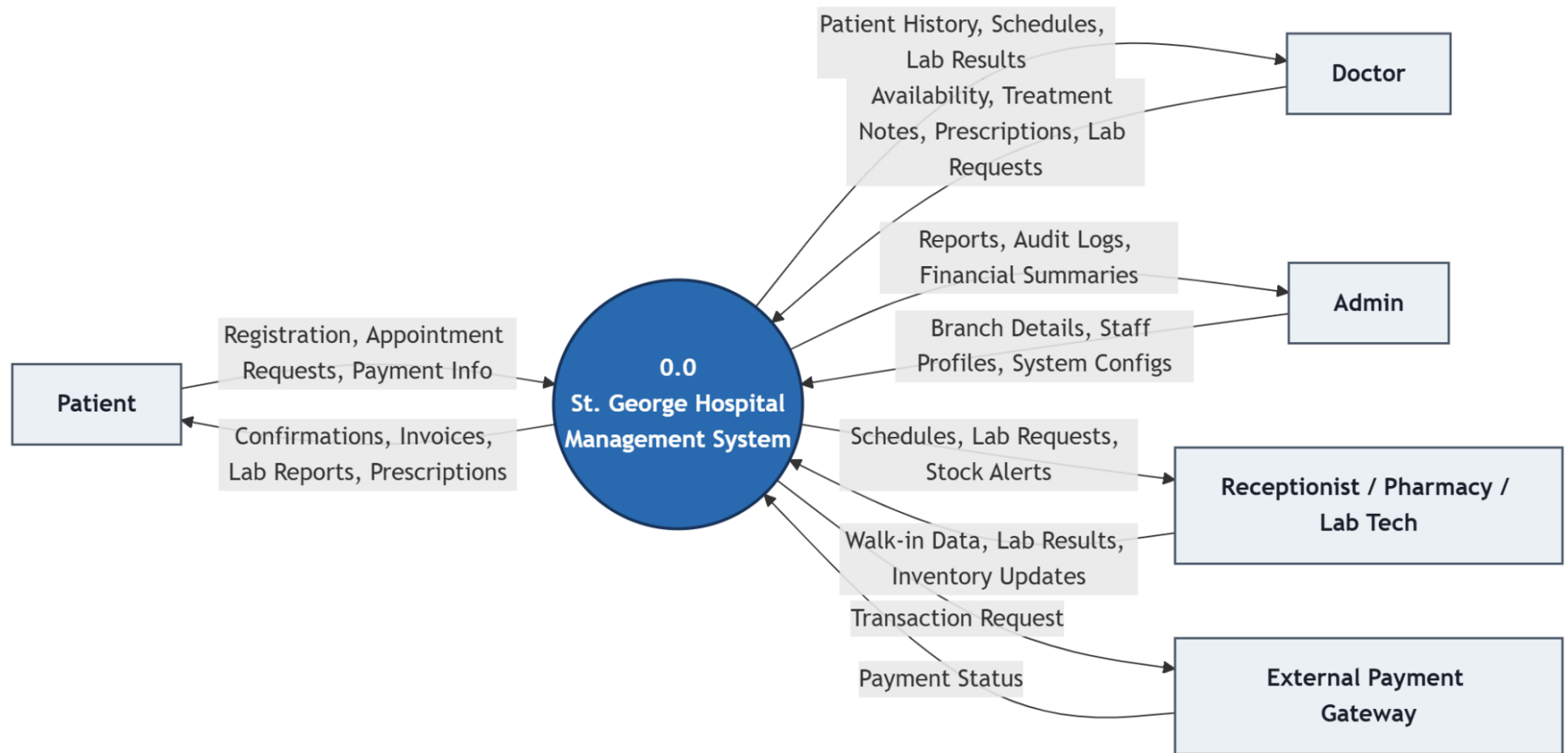


Figure F4. Level 0 data-flow diagram.

1DFD

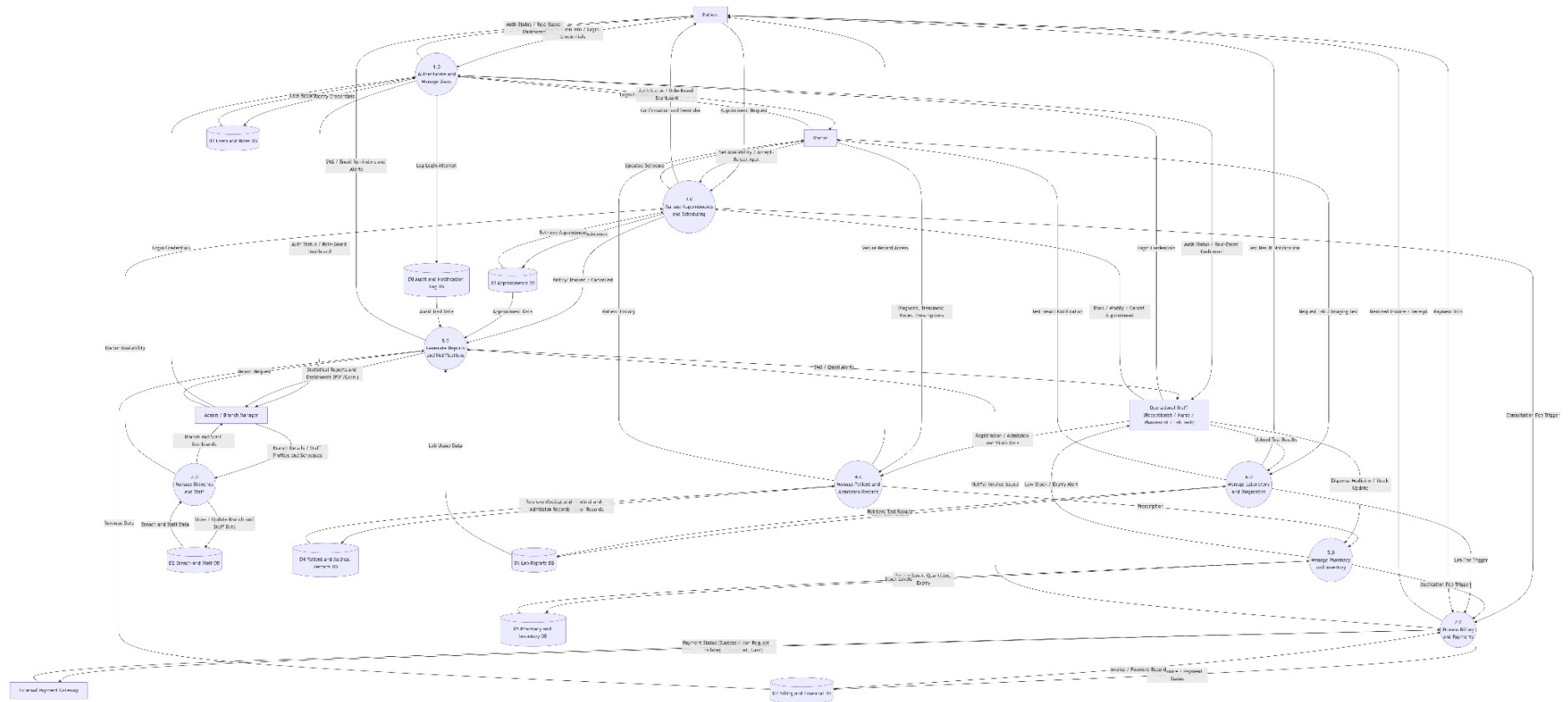


Figure F5. Level 1 data-flow diagram (redraw required for final readability).

Sequence Diagram

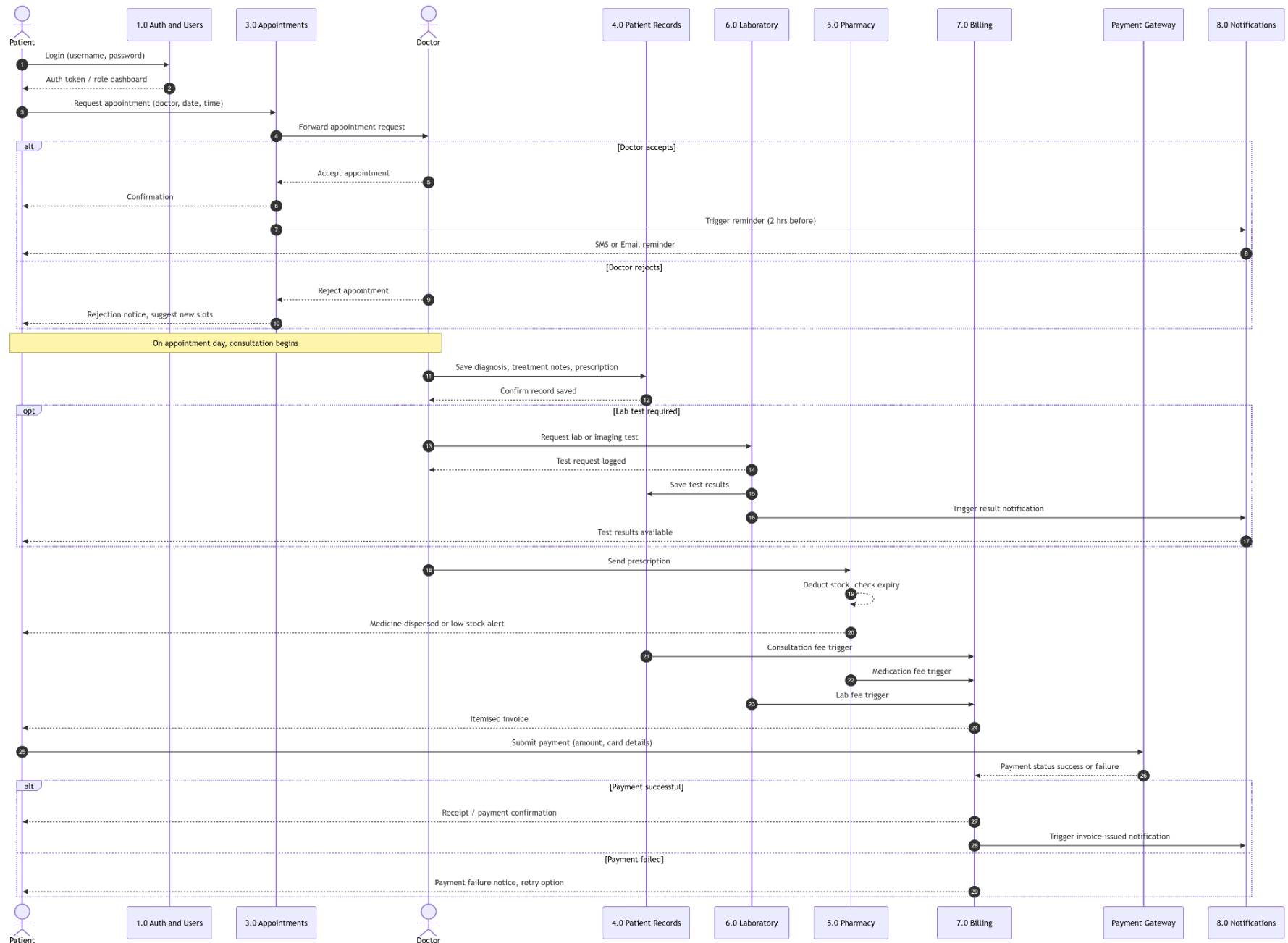


Figure F6. Supplied end-to-end sequence draft.

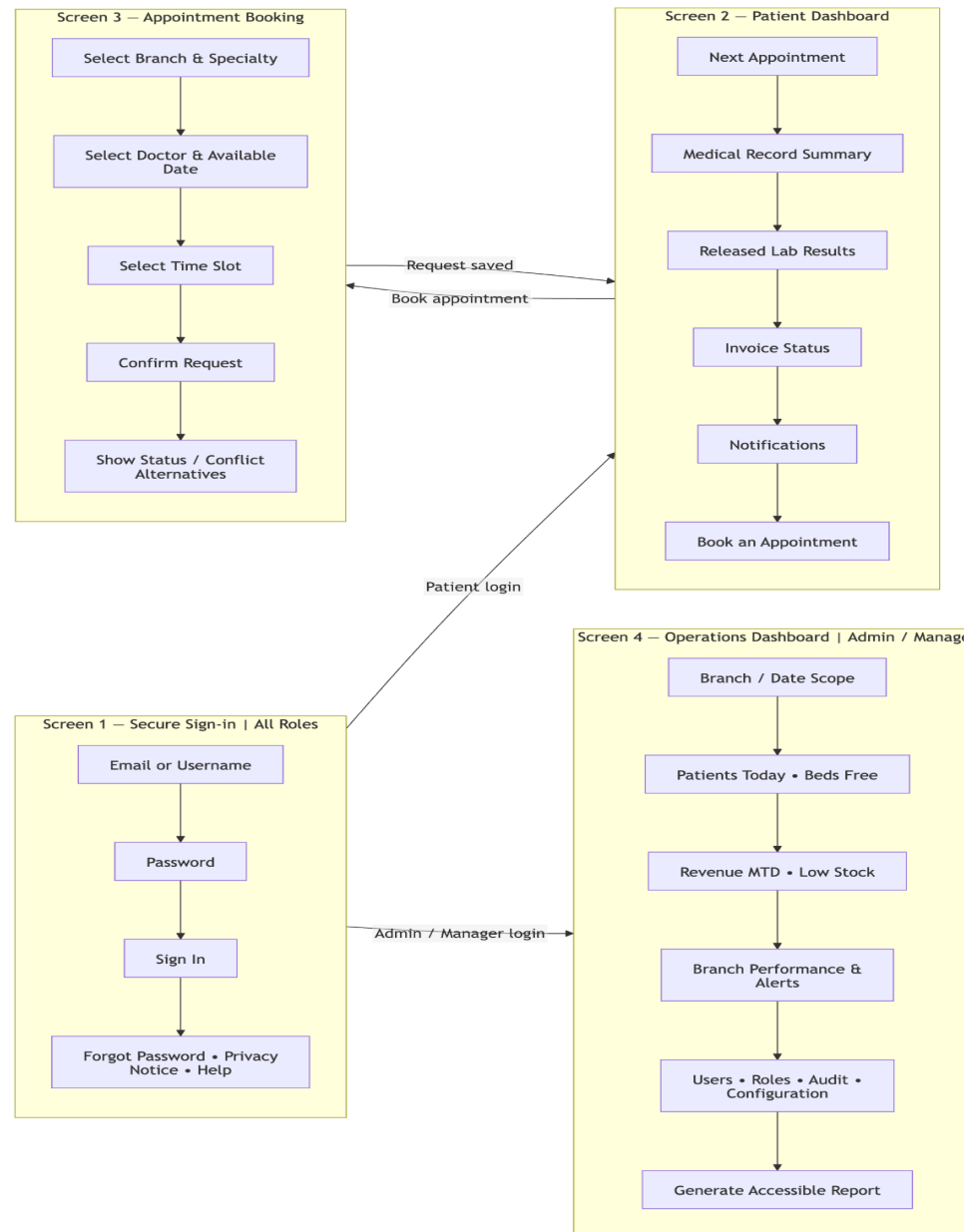
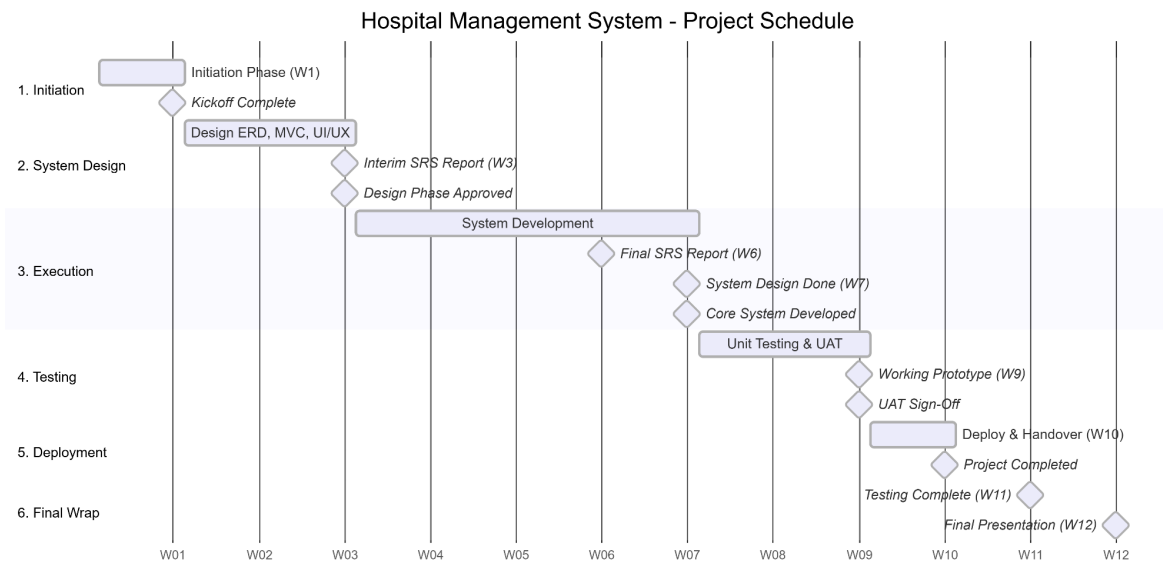


Figure F7. Proposed UI storyboard for principal user journeys.

Gantt Chart



Appendix G — Individual contribution statement

Member	Assigned deliverables
Priya Paneru	Schedule, scope, risks, decisions, submission
Usha Tamang	Requirements, traceability, use cases, DFD
Manish Das	PHP/API, database, deployment/security
Oyndrila Oyshi Saha	UI/UX, accessibility, frontend
Krishala Niroula	Test cases, defects, regression/UAT

